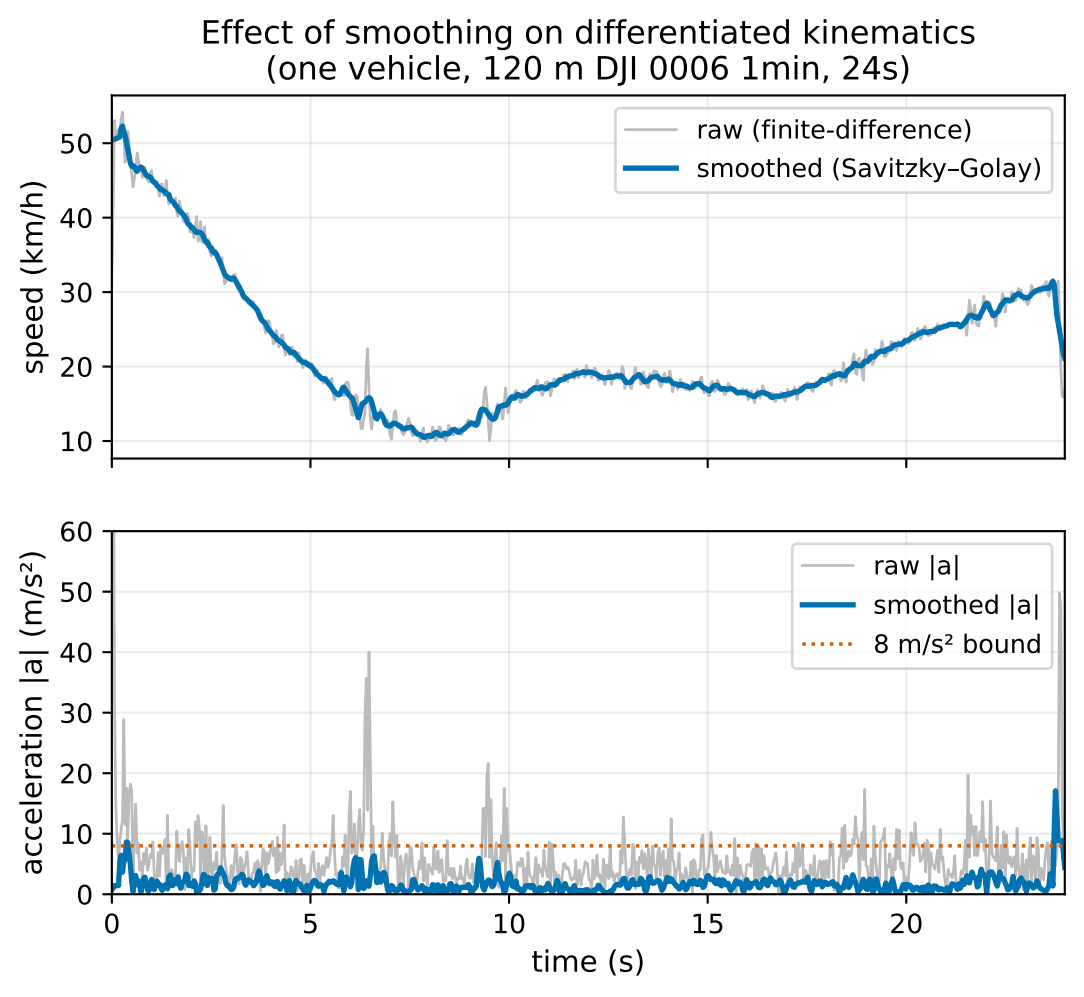
Effect of smoothing on differentiated kinematics
(one vehicle, 120 m DJI 0006 1min, 24s)
raw (finite-difference)
smoothed (Savitzky–Golay)
speed (km/h)
50
40
30
20
10
raw |a|
smoothed |a|
8 m/s² bound
acceleration |a| (m/s²)
60
50
40
30
20
10
0
0
5
10
15
20
time (s)